Homework 2

Problem 1

Virtual memory uses a page table to track the mapping of virtual addresses to physical addresses. This exercise shows how this table must be updated as addresses are accessed. The following table is a stream of virtual addresses as seen on a system. Assume 4 KiB pages, a 4-entry fully associative TLB, and true LRU replacement. If pages must be brought in from disk, increment the next largest page number.

4095, 31272, 15789, 15000, 7193, 4096, 8912

TLB

Valid	Tag	Physical Page Number
1	11	12
1	7	4
1	3	6
0	4	9

Page Table

Valid	Physical page or in disk
1	5
0	Disk
0	Disk
1	6
1	9
1	11
0	Disk
1	4
0	Disk
0	Disk
1	3
1	12

a) Given the address stream in the table, and the initial TLB and page table states shown above, show the final state of the system. Also list for each reference if it is a hit in the TLB, a hit in the page table, or a page fault.
b) Repeat Exercise a), but this time use 16 KiB pages instead of 4 KiB pages. What would be some of the advantages of having a larger page size? What are some of the disadvantages?
c) Show the final contents of the TLB if it is 2-way set associative. Also show the contents of the TLB if it is direct mapped. Discuss the importance of having a TLB to high performance. How would virtual memory accesses be handled if there were no TLB?

There are several parameters that impact the overall size of the page table. Listed below are

several key page table parameters.

Virtual address size	Page size	Page table entry size
32 bits	4 KiB	8 bytes

d) Given the parameters in the table above, calculate the total page table size for a system running 5 applications that utilize half of the memory available.
e) Given the parameters in the table above, calculate the total page table size for a system running 5 applications that utilize half of the memory available, given a two-level page table approach with 256 entries. Assume each entry of the main page table is 6 bytes. Calculate the minimum and maximum amount of memory required.
f) A cache designer wants to increase the size of a 4 KiB virtually indexed, physically tagged cache. Given the page size listed in the table above, is it possible to make a 16 KiB direct-mapped cache, assuming 2 words per block? How would the designer increase the data size of the cache?

Problem 2

Consider a system with 40-bit virtual addresses, 36-bit physical addresses, and 64 KiB (2^{16} bytes) pages. The system uses a page map to translate virtual addresses to physical addresses; each page map entry includes dirty (D) and resident (R) bits.
a) Assuming a flat page map, what is the size of each page map entry, and how many entries does the page map have?

Size of page map entry in bits: _____

Number of entries in the page map: _____

b) If changed the system to use 16 KiB (2^{14} bytes) pages instead of 64 KiB pages, how would the number of entries in the page map change? Please give the ratio of the new size to the old size.

(# entries with 16 KiB pages) / (# entries with 64 KiB pages): _____

c) Assume we still use 64 KiB pages. The contents of the page map and TLB are shown to the right. The page map uses an LRU replacement policy and the LRU page (shown below) will be chosen for replacement. For each of these four accesses, compute its corresponding physical address and indicate whether the access causes a TLB miss and/or a page fault. Assume each access starts with the TLB and Page Map state shown to the right.

TLB

VPN (tag)	V	D	PPN
0x0	1	0	0xBE7A
0x3	0	0	0x7
0x5	1	1	0xFF
0x2	1	0	0x900

Fill in table below

	Virtual Addr	PPN (in hex)	PhysAddr (in Hex)	TLB Miss? (Y/N)	Page Fault? (Y/N)
1.	**0x06004**				
2.	**0x30286**				
3.	**0x68030**				
4.	**0x4BEEF**				

Page Map

VPN	R	D	PPN	
0	1	0	0xBE7A	
1	0	0	---	
2	1	0	0x900	
3	1	0	0x8	⇦ LRU PAGE
4	0	0	---	
5	1	1	0xFF	
6	1	0	0x70	

Problem 3

For a direct-mapped cache design with a 32bit address, the following bits of the address are used to access the cache.

Tag	Index	Offset
31-13	12-6	5-0

a) What is the cache line size(in words)?
b) How many entries does the cache have?
c) What is the ratio between total bits required for such a cache implementation over the data storage bits? (Note the structure of the direct-mapped cache! Don't forget the valid bit.)

Problem 4

Cache is important to providing a high-performance memory hierarchy to processors. Below is a list of 32-bit memory address references, **given as word address**.
3, 180, 43, 2, 181, 88, 164, 180, 89.

(a) For each of these references, identify the binary address, the tag, and the index given a direct-mapped cache with 8 two-word blocks. Also list if each reference is a hit or a miss, assuming the cache is initially empty.

Word address	Binary address	Tag	Index	Hit/Miss
3	0000 0011			
180	1011 0100			
43				
2				
181				
88				
164				
180				
89				

(b) For each of these references, identify the binary address, the tag, and the index given a direct-mapped cache with 4 four-word blocks. Also list if each reference is a hit or a miss, assuming the cache is initially empty.

Word address	Binary address	Tag	Index	Hit/Miss
3	0000 0011			
180	1011 0100			
43				

2				
181				
88				
164				
180				
89				

Problem 5

This exercise examines the impact of cache designs. For this exercise, the table of **word address** streams is shown in below.

1, 134, 212, 1, 135, 213, 162, 161, 2, 44, 41, 221

What is the miss rate for a fully associative cache with two-word blocks and a total size of 8 words, using LRU replacement?

Problem 6

You are trying to reverse-engineer the characteristics of a cache in a system, so that you can design a more efficient, machine-specific implementation of an algorithm you are working on. To do so, you have come up with three patterns that access various bytes in the system in an attempt to determine the following four cache characteristics:

• Cache block size (8, 16, 32, 64, or 128 B)
• Cache associativity (1-, 2-, 4-, or 8-way)
• Cache size (4 or 8 KB)
• Cache replacement policy (LRU or FIFO)

However, the only statistic that you can collect on this system is cache hit rate after performing the
access pattern. Here is what you observe (the address is in bytes):

Sequence	Addresses Accessed (Oldest → Youngest)								Hit Rate
1.	0	4	8	16	64	128			1/2
2.	31	8192	63	16384	4096	8192	64	16384	5/8
3.	32768	0	129	1024	3072	8192			1/3

Assume that the cache is initially empty at the beginning of the first sequence, but not at the
beginning of the second and third sequences. The sequences are executed back-to-back, i.e., no other
accesses take place between the three sequences. Thus, at the beginning of the second (third)
sequence, the contents are the same as at the end of the first (second) sequence.

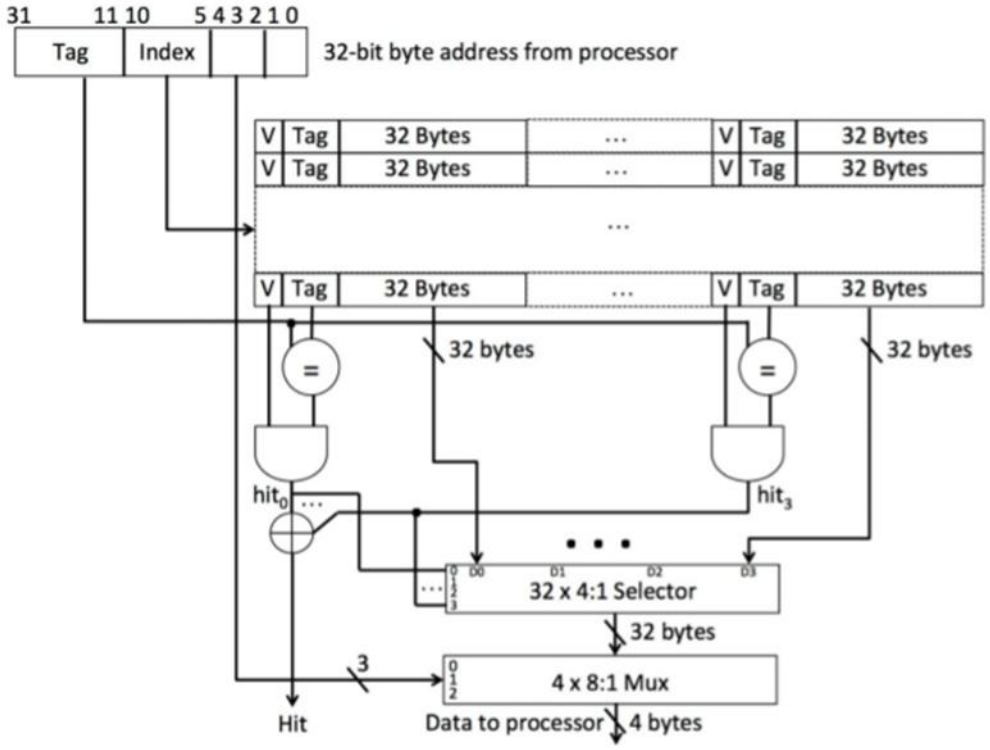
Based on what you observe, what are the following characteristics of the cache?

Explain the reason.

(a) Cache block size (8, 16, 32, 64, or 128 B)?
(b) Cache associativity (1-, 2-, 4-, or 8-way)?
(c) Cache size (4 or 8 KB)?
(d) Cache replacement policy (LRU or FIFO)?

Problem 7

You are given the sketch of a cache design below:



Note: a 4:1 Selector has four control inputs labeled 0, 1, 2, 3 and four data inputs D0, D1, D2, D3. It connects Di to the output when the hit control signal is true. Assume that at most one control input is true at any time. The selector function is undefined when none of the control inputs are true.

Answer the following questions about the cache above.

a) What is the block size of the cache in bytes?
b) What is the number of blocks in this cache?
c) What is the total data capacity of the cache in bytes?
d) What is the associativity of the cache?
e) Is this a write-through or write-back cache?
f) What is the total number of valid bits in the cache?
g) What is the total number of tag bits in the cache?

Problem 8

Caches are important to providing a high performance memory hierarchy to process. Below is a list of 32bit memory address references, **given as word address.**
7, 215, 175, 215, 7, 84, 65, 174, 64, 105, 85, 215

a) You're asked to optimize a cache design for the given references. There are three direct-mapped cache designs possible, all with total of 8 words of data:
C1 has 1-word blocks, C2 has 2-word blocks, and C3 has 4-word blocks.

In term of miss rate, which cache design is the best? If the miss stall time is 25 cycles, and C1 has an access time of 2 cycles, C2 takes 3 cycles, and C3 takes 5 cycles, which is the best cache design?